

CS471 Project 3

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1 INTRODUCTION

This report contains the results of running 30 iterations of Differential Evolution using 10 different strategies and Particle Swarm Optimization on 10 different fitness functions in 10, 20, and 30 dimensions. For each iteration the best fitness is saved. The description of the benchmark functions are listed in Section 2. The experimentation results can be found in Section 3. Analysis of the experimentation is found in Section 4.

2 BENCHMARK FUNCTION DESCRIPTION AND FULL RESULTS

Each of the following functions were evaluated 30 times using Differential Evolution (using its 10 strategies) and Particle Swarm Optimization in 10, 20, and 30 dimensions. The Experiment specifications can be found in Table 2.1. The results can be seen in Section 3.

2.1 BENCHMARKING FUNCTIONS

1. Schwefel's function

$$f_1(x) = 418.9829 * n - \sum_{i=1}^n -x_i \cdot \sin(\sqrt{|x_i|}) \quad (2.1)$$

2. 1st De Jong's function:

$$f_2(x) = \sum_{i=1}^n x_i^2 \quad (2.2)$$

3. Rosenbrock:

$$f_3(x) = \sum_{i=1}^{n-1} 100(x_i^2 - x_{i+1})^2 \quad (2.3)$$

4. Rastrigin:

$$f_4 = 10 * n \sum_{i=1}^n (x_i^2 - 10 \cdot \cos(2\pi \cdot x_i)) \quad (2.4)$$

5. Griewangk:

$$f_5 = 1 + \sum_{i=1}^n \frac{x_i^2}{4000} - \prod_{i=1}^n \cos\left(\frac{x_i}{\sqrt{i}}\right) \quad (2.5)$$

6. Sine Envelope Sine Wave:

$$f_6 = - \sum_{i=1}^{n-1} 0.5 + \frac{\sin(x_i^2 + x_{i+1}^2 - 0.5)}{(1 + 0.001(x_i^2 + x_{i+1}^2))^2} \quad (2.6)$$

7. Stretched V Sine Wave:

$$f_7 = \sum_{i=1}^{n-1} \left(\sqrt[4]{x_i^2 + x_{i+1}^2} \cdot \sin\left(50 \sqrt[10]{x_i^2 + x_{i+1}^2}\right)^2 + 1 \right) \quad (2.7)$$

8. Ackley's One:

$$f_8 = \sum_{i=1}^{n-1} \frac{1}{e^{0.2}} \sqrt{x_i^2 + x_{i+1}^2} + (\cos(2x_i) + \sin(2x_{i+1})) \quad (2.8)$$

9. Ackley's Two:

$$f_9 = \sum_{i=1}^{n-1} 20 + e - \frac{20}{e^{0.2 \sqrt{\frac{x_i^2 + x_{i+1}^2}{2}}}} - e^{0.5(\cos(2\pi \cdot x_i) + \cos(2\pi \cdot x_{i+1}))} \quad (2.9)$$

10. Egg Holder:

$$f_{10} = \sum_{i=1}^{n-1} -x_i \cdot \sin\left(\sqrt{|x_i - x_{i+1} + 1 - 47|}\right) - (x_{i+1} + 47) \cdot \sin\left(\sqrt{\left|x_{i+1} + 46 \frac{x_i}{2}\right|}\right) \quad (2.10)$$

Table 2.1: Experiment Specifications

	Name	Dimensions	Range	Experimentations	Generations	Population
f_1	Schwefel	10,20,30	$[-512,512]^n$	30	100	200
f_2	De Jong 1	10,20,30	$[-100,100]^n$	30	100	200
f_3	Rosenbrock's Saddle	10,20,30	$[-100,100]^n$	30	100	200
f_4	Rastrigin	10,20,30	$[-30,30]^n$	30	100	200
f_5	Griewangk	10,20,30	$[-500,500]^n$	30	100	200
f_6	Sine Envelope Sine Wave	10,20,30	$[-30,30]^n$	30	100	200
f_7	Stretch V Sine Wave	10,20,30	$[-30,30]^n$	30	100	200
f_8	Ackley One	10,20,30	$[-32,32]^n$	30	100	200
f_9	Ackley Two	10,20,30	$[-32,32]^n$	30	100	200
f_{10}	Egg Holder	10,20,30	$[-500,500]^n$	30	100	200

2.2 EXPERIMENT SPECIFICATIONS

Specifications for each experiment specifies which function to evaluate fitness with, how many dimensions to evaluate on, the Domain of the search space, the number of experimentations, number of generations, the population size for each generation, can be seen in Table 2.1.

3 EXPERIMENTATION

3.1 DIFFERENTIAL EVOLUTION EXPERIMENTATION

The results for Differential Evolution in 10 Dimensions for 10 different strategies can be seen in tables 3.1, 3.2, and 3.3 with each functions best average across the 10 strategies in bold. The results for Differential Evolution in 30 Dimensions for 10 different strategies can be seen in tables 3.4, 3.5, and 3.6 with each functions best average across the 10 strategies in bold. The results for Differential Evolution in 30 Dimensions for 10 different strategies can be seen in tables 3.7, 3.8, and 3.9 with each functions best average across the 10 strategies in bold. In 10 Dimensions the best strategy was strategy 1(DE/best/1/exp) which had the best averages for 6 of the fitness functions. In 20 Dimensions the best strategy was a tie between strategy 1 and strategy 3(DE/rand-to-best/1/exp) both having the best averages for 4 of the 10 fitness functions. In 30 Dimensions the best strategy was strategy 3 with 7 of the best averages, while strategy 1 held the best average for Egg Holder.

3.2 PARTICLE SWARM OPTIMIZATION EXPERIMENTATION

The results for Particle Swarm Optimization can be seen in table 3.10.

4 ANALYSIS

The Comparison of Differential Evolutions best strategy for each fitness function and Particle Swarm Optimization can be seen in table 4.1 (10 Dimensions), 4.2 (20 Dimensions), and 4.3 (30 Dimensions). In 10 dimensions the split between Differential Evolution's Best strategy and Particle Swarm Optimization finding a lower fitness value is 6:4. In 20 dimensions the split is 3:7. In 30 dimensions the split is 2:8. This shows that as the dimensions increase Particle Swarm Optimization will in general outperform Differential Evolution given the same number of generations. Overall the Standard Deviation of Differential Evolution is smaller than Particle Swarm Optimization, this may be due to Particle Swarm Optimization converging rapidly in some of the iterations.

Table 3.1: DE Results for 10 Dimensions Using Strategies 1-4

Problem	DE/best/1/exp			DE/rand/1/exp			DE/rand-to-best/1/exp			DE/best/2/exp		
	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)
f_1	109.19	72.61	1.20	204.03	48.16	1.21	587.19	72.19	1.22	314.68	73.73	1.24
f_2	0.00	0.00	0.88	1.88	0.44	0.89	0.00	0.00	0.89	1.48	0.40	0.92
f_3	2082.55	941.41	0.92	26366.81	11387.54	0.93	1819.22	905.92	0.92	137707.33	80841.33	0.94
f_4	-8014.30	416.00	1.15	-6164.13	521.61	1.16	-7718.45	536.22	1.17	-4128.31	674.61	1.18
f_5	0.81	0.09	1.20	1.15	0.06	1.22	0.79	0.12	1.23	1.45	0.14	1.25
f_6	-3.43	0.15	1.1	-3.31	0.16	1.20	-3.45	0.11	1.18	-3.16	0.18	91.22
f_7	9.71	0.22	1.97	9.63	0.24	1.99	9.70	0.15	1.98	9.71	0.29	2.00
f_8	-23.80	1.15	1.45	-17.55	2.24	1.47	-21.57	1.09	1.45	-9.73	3.36	1.49
f_9	10.17	1.84	2.33	25.42	2.66	2.34	11.05	1.76	2.34	42.74	4.73	2.32
f_{10}	-6401.91	295.80	1.52	-5997.70	205.64	1.54	-5976.82	230.43	1.52	-5900.62	193.95	1.55

[†] Run on Intel Core i5 2.9 GHz, 16 GB RAM

Table 3.2: DE Results for 10 Dimensions Using Strategies 5-8

Problem	DE/rand/2/exp			DE/best/1/bin			DE/rand/1/bin			DE/rand-to-best/1/bin		
	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)
f_1	511.79	122.69	2.25	511.97	138.69	1.74	446.11	65.53	1.78	788.35	63.91	1.78
f_2	0.00	0.00	1.93	0.00	0.00	1.41	2.69	0.82	1.43	0.00	0.00	1.46
f_3	8361.63	3683.27	1.94	10730.15	5124.20	1.45	1098249.34	732180.32	1.47	6616.41	3174.01	1.47
f_4	-5934.31	814.41	2.17	-5860.50	734.93	1.70	-1425.79	1383.02	1.70	-5969.78	602.83	1.71
f_5	0.94	0.09	2.24	0.96	0.07	1.74	2.46	0.39	1.77	0.83	0.13	1.77
f_6	-2.54	0.18	2.21	-2.54	0.14	1.72	-2.35	0.13	1.74	-2.61	0.18	1.75
f_7	11.56	0.58	3.00	11.45	0.76	2.52	11.50	0.68	2.53	11.49	0.48	2.54
f_8	-15.41	2.86	2.48	-14.90	2.67	2.00	0.58	3.70	2.02	-13.89	2.31	2.02
f_9	18.67	3.34	3.36	18.15	2.88	2.87	61.11	4.94	2.83	19.73	2.97	2.89
f_{10}	-6099.05	487.38	2.54	-5968.76	584.76	2.07	-5338.18	253.82	2.08	-5382.90	457.67	2.10

[†] Run on Intel Core i5 2.9 GHz, 16 GB RAM

Table 3.3: DE Results for 10 Dimensions Using Strategies 9 and 10

Problem	DE/best/2/bin			DE/rand/2/bin		
	Avg	Std	T(s)	Avg	Std	T(s)
f_1	612.94	93.14	1.81	660.64	55.60	1.83
f_2	0.96	0.34	1.47	120.43	26.35	1.48
f_3	16008182.23	8458541.35	1.51	57732730.98	1.53	1.77
f_4	8132.46	3231.80	1.74	15121.85	4223.53	1.77
f_5	7.00	1.72	1.81	13.23	3.24	1.83
f_6	-2.31	0.22	1.80	-2.20	0.20	1.82
f_7	11.26	0.69	2.58	11.53	0.78	2.59
f_8	20.56	4.83	2.05	30.59	5.07	2.08
f_9	93.61	7.86	2.85	105.56	7.09	2.86
f_{10}	-5325.92	297.50	2.14	-5125.25	231.08	2.15

[†] Run on Intel Core i5 2.9 GHz, 16 GB RAM

Table 3.4: DE Results for 20 Dimensions Using Strategies 1-4

Problem	DE/best/1/exp			DE/rand/1/exp			DE/rand-to-best/1/exp			DE/best/2/exp		
	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)
f_1	1458.93	125.56	1.72	1509.17	90.34	1.74	2006.27	142.56	1.73	1692.29	109.11	1.81
f_2	51.69	10.59	1.08	444.21	79.21	1.10	40.31	8.32	1.11	568.78	98.49	1.16
f_3	8560777.68	3287262.21	1.17	36643265.57	14689839.38	1.18	7572198.50	3002752.10	1.19	143029866.33	57014939.28	1.24
f_4	3275.88	3755.14	1.63	23439.07	6613.63	1.65	4252.45	3004.21	1.80	56624.61	13049.42	1.83
f_5	5.62	0.73	1.75	10.80	1.97	1.76	5.08	0.80	1.80	21.09	3.98	1.83
f_6	-5.27	0.31	1.72	-5.02	0.25	1.74	-5.20	0.24	1.79	-4.67	0.25	1.82
f_7	24.69	0.79	3.39	24.59	0.87	3.39	24.40	0.82	3.52	25.20	1.18	2.32
f_8	14.07	5.61	2.23	33.74	5.41	2.24	13.79	5.15	2.30	62.24	8.57	2.32
f_9	130.63	9.10	4.08	169.73	11.80	4.11	126.35	6.79	4.20	221.10	13.98	4.17
f_{10}	-10339.85	557.57	2.43	-9970.72	405.42	2.47	-9929.16	404.64	2.51	-9904.72	336.48	2.55

[†] Run on Intel Core i5 2.9 GHz, 16 GB RAM

Table 3.5: DE Results for 20 Dimensions Using Strategies 5-8

Problem	DE/rand/2/exp			DE/best/1/bin			DE/rand/1/bin			DE/rand-to-best/1/bin		
	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)
f_1	1609.66	119.86	3.61	1605.65	149.36	3.03	1979.77	97.78	3.11	2744.09	286.60	3.19
f_2	0.05	0.02	2.95	0.04	0.02	2.39	1253.57	254.05	2.46	0.01	0.00	2.50
f_3	320084576.27	121236568.77	3.01	314443973.87	12458804.44	2.79	2066867174.40	635089963.91	2.52	231905058.93	93507057.49	2.62
f_4	60998.14	10501.22	3.51	63092.12	11261.81	2.95	214501.97	45052.47		46430.94	11932.17	3.08
f_5	23.78	4.69	3.65	22.49	4.58	3.15	74.52	11.89	3.11	18.16	3.26	3.21
f_6	-3.24	0.21	3.64	-3.27	0.25	3.07	-3.08	0.21	3.10	-3.28	0.25	3.21
f_7	35.39	2.06	5.39	36.25	1.82	5.02	36.78	1.82	4.76	34.75	1.69	3.60
f_8	70.95	10.62	4.09	69.88	13.66	3.67	136.18	11.76	3.66	64.56	10.48	3.60
f_9	220.75	14.08	6.21	220.50	3.67	5.88	275.11	11.31	5.42	232.87	12.89	5.43
f_{10}	-8576.13	913.57	4.64	-8501.69	1115.35	3.76	-7477.70	396.01	3.83	-77722.49	813.81	3.83

¹ Run on Intel Core i5 2.9 GHz, 16 GB RAM

Table 3.6: DE Results for 20 Dimensions Using Strategies 9 and 10

Problem	DE/best/2/bin			DE/rand/2/bin		
	Avg	Std	T(s)	Avg	Std	T(s)
f_1	1941.38	96.01	3.15	2275.38	117.99	3.16
f_2	1504.13	375.91	2.48	9625.95	1223.33	2.49
f_3	651924882.13	1866447972.13	2.59	7834521523.20	2261482347.20	2.61
f_4	414727.94	56737.06	3.06	463895.07	53538.37	3.07
f_5	143.07	21.62	3.22	169.41	19.90	3.18
f_6	02.82	0.22	3.17	-2.74	0.20	3.18
f_7	38.45	2.80	4.81	38.15	1.96	4.81
f_8	193.70	14.11	3.66	214.45	18.36	3.68
f_9	297.74	8.42	5.35	305.01	8.55	5.35
f_{10}	-7798.96	538.50	3.86	-7588.45	437.41	3.89

[†] Run on Intel Core i5 2.9 GHz, 16 GB RAM

Table 3.7: DE Results for 30 Dimensions Using Strategies 1-4

Problem	DE/best/1/exp			DE/rand/1/exp			DE/rand-to-best/1/exp			DE/best/2/exp		
	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)
f_1	3291.38	307.29	2.25	3174.85	189.10	2.27	4085.93	305.56	2.27	3271.05	228.65	2.31
f_2	1130.02	158.15	1.31	3550.13	520.64	1.31	922.37	134.89	1.31	4365.92	635.23	1.34
f_3	322113201.33	105145134.82	1.43	678758640.00	236612241.61	1.43	267732335.47	70431483.991	1.43	1747507485.87	539545387.43	1.46
f_4	144856.74	27033.67	2.11	226465.24	25536.36	2.13	127666.42	23224.61	2.12	358752.93	59314.97	2.15
f_5	34.45	5.16	2.29	50.50	7.22	2.31	33.83	4.38	2.31	85.54	12.14	2.34
f_6	-6.57	0.26	2.27	-6.34	0.34	2.29	-6.68	0.25	2.27	-5.96	0.37	2.32
f_7	43.39	1.61	4.81	43.07	2.00	4.81	42.28	1.62	4.83	45.34	1.84	4.83
f_8	107.93	12.69	3.00	130.41	15.16	3.03	101.70	7.37	3.02	187.35	15.81	3.05
f_9	328.31	11.88	5.78	361.93	15.51	5.78	323.25	10.95	5.81	411.94	10.07	5.76
f_{10}	-13302.85	473.00	3.36	-12946.88	516.38	3.38	-12919.62	585.65	3.37	-13118.74	481.49	3.41

¹ Run on Intel Core i5 2.9 GHz, 16 GB RAM

Table 3.8: DE Results for 30 Dimensions Using Strategies 5-8

Problem	DE/rand/2/exp			DE/best/1/bin			DE/rand/1/bin			DE/rand-to-best/1/bin		
	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)
f_1	3032.87	515.36	4.82	2910.27	212.40	4.34	3980.44	299.99	4.38	5044.02	456.77	4.43
f_2	5.56	2.60	3.85	5.37	2.28	3.38	9950.65	1248.73	3.39	0.66	0.17	3.46
f_3	5701006873.60	1726947190.56	3.99	5692389546.67	1559733959.77	3.50	15342383684.27	2908282940.43	3.55	4288837879.47	1141914035.03	3.59
f_4	558198.66	78172.86	4.67	564790.86	83131.31	4.18	1102482.75	114516.63	4.22	466696.44	68821.56	4.25
f_5	126.76	17.34	4.85	129.10	17.95	4.38	246.24	29.70	4.42	109.81	12.10	4.47
f_6	-3.85	0.30	4.86	-3.75	0.26	4.37	-3.53	0.28	4.40	-3.89	0.31	4.45
f_7	64.55	2.30	7.40	64.09	2.70	6.92	65.62	3.28	6.93	63.65	2.59	7.00
f_8	231.10	20.16	5.59	229.07	21.50	5.25	328.89	24.06	5.16	218.60	14.86	5.19
f_9	437.41	11.76	8.30	436.59	13.32	7.81	473.15	9.27	7.80	440.87	12.83	7.96
f_{10}	-9594.32	1255.20	5.95	-9240.16	838.63	5.47	-8989.50	776.04	5.50	-8967.80	957.30	5.55

[†] Run on Intel Core i5 2.9 GHz, 16 GB RAM

Table 3.9: DE Results for 30 Dimensions Using Strategies 9 and 10

Problem	DE/best/2/bin			DE/rand/2/bin		
	Avg	Std	T(s)	Avg	Std	T(s)
f_1	3836.83	381.54	4.48	4573.00	323.63	4.50
f_2	13659.99	1845.83	3.48	35137.98	3300.20	3.52
f_3	22610254438.40	4882364456.91	3.65	23801192379.73	3223346820.68	3.69
f_4	1501080.81	162142.52	4.32	1478698.96	169323.22	4.34
f_5	341.09	33.36	4.52	357.21	29.91	4.54
f_6	-3.28	0.29	4.51	-3.22	0.26	4.54
f_7	68.23	2.23	7.01	68.76	2.80	7.03
f_8	392.74	22.20	5.26	402.52	21.30	5.29
f_9	493.54	9.41	7.83	498.67	9.71	7.82
f_{10}	-9184.97	863.90	5.60	-8704.86	831.57	5.61

[†] Run on Intel Core i5 2.9 GHz, 16 GB RAM

Table 3.10: PSO results

Problem	PSO (10 dim)			PSO (20 dim)			PSO (30 dim)		
	Avg	Std	T(s)	Avg	Std	T(s)	Avg	Std	T(s)
f_1	969.80	193.92	2.17	2130.82	300.46	4.33	3419.73	460.37	6.43
f_2	0.00	0.00	1.85	0.00	0.00	3.70	0.02	0.02	5.50
f_3	68099.70	253343.06	1.89	67555.95	253483.32	3.76	35293.21	182258.68	5.61
f_4	-8727.93	547.66	2.12	-25241.43	5229.42	4.20	-40087.53	18011.75	6.29
f_5	0.14	0.06	2.19	0.03	0.03	4.33	0.04	0.05	6.48
f_6	-2.96	0.23	2.14	-4.06	0.53	4.31	-4.73	0.53	6.44
f_7	9.02	0.09	2.94	19.71	1.08	5.98	32.15	2.97	9.03
f_8	-19.97	6.67	2.44	-4.89	20.16	4.85	47.37	35.60	7.22
f_9	30.93	25.68	3.25	138.23	37.20	6.62	273.96	73.32	9.96
f_{10}	-5910.74	664.85	2.50	-10710.97	1191.94	5.03	-14420.21	1386.45	7.78

[†] Run on Intel Core i5 2.9 GHz, 16 GB RAM

Table 4.1: Analysis for 10 Dimensions

Problem	DE	PSO	t-value	p-value	$p < 0.05$
Schwefel	109.19	969.80	-21.18	0.00	yes
1 De Jong	0.00	0.00	2.97	0.01	yes
Rosenbrock	1819.22	68099.70	-1.43	0.16	no
Rastrign	-8014.30	-8727.93	5.80	0.00	yes
Griewangk	0.79	0.14	26.20	0.00	yes
Sine Envelope	-3.45	-2.96	-10.82	0.00	yes
Stretched V	9.63	9.02	13.63	0.00	yes
Ackley One	-23.80	-19.97	-3.20	0.00	yes
Ackley Two	10.17	30.93	-4.42	0.00	yes
Egg Holder	-6401.91	-5910.74	-3.50	0.00	yes

Table 4.2: Analysis for 20 Dimensions

Problem	DE	PSO	t-value	p-value	$p < 0.05$
Schwefel	1458.93	2130.82	-10.98	0.00	yes
1 De Jong	0.01	0.00	13.73	0.00	yes
Rosenbrock	7572198.50	67555.95	13.68	0.00	yes
Rastrign	3275.88	-25241.43	22.82	0.00	yes
Griewangk	5.08	0.03	34.56	0.00	yes
Sine Envelope	-5.27	-4.06	-11.77	0.00	yes
Stretched V	24.40	19.71	18.62	0.00	yes
Ackley One	13.79	138.23	5.13	0.00	yes
Ackley Two	126.35	-10710.97	-1.87	0.07	no
Egg Holder	-13302.85	-10339.85	1.51	0.14	no

Table 4.3: Analysis for 30 Dimensions

Problem	DE	PSO	t-value	p-value	$p < 0.05$
Schwefel	2910.27	3419.73	-5.17	0.00	yes
1 De Jong	0.66	0.02	19.13	0.00	yes
Rosenbrock	267732335.47	35293.21	20.81	0.00	yes
Rastrign	127666.42	40087.53	30.27	0.00	yes
Griewangk	1 33.83	0.04	42.24	0.00	yes
Sine Envelope	-6.68	-4.73	-16.05	0.00	yes
Stretched V	42.28	32.15	18.39	0.00	yes
Ackley One	101.70	47.37	7.86	0.00	yes
Ackley Two	323.25	273.96	3.62	0.00	yes
Egg Holder	-13302.85	-14420.21	3.84	0.00	yes